

Advanced Computing Project

Projeto em Computação Avançada (MEI/MCA)

Roadmap

University of Minho
2025-2026



Assignment

❖ Practical assignment (no written exam!)

- Research-oriented projects (challenge -> solution)
 1. Survey related work (the problem might not be new)
 2. Design and prototype implementation (overcome limitations of prior work)
 3. Experimental evaluation (validate the performance of the proposed solution)
 4. Paper writing

❖ ECTS

- MCA - 10 ECTS
- MEI - 5 ECTS

Class organization

❖ Tutorial classes

- **Tutorial 1:** *How to search and analyze related work?* (Week 3: February 20th)
- **Tutorial 2:** *How to evaluate systems?* (Week 6: March 13th)
- **Tutorial 3:** *How to write (and review) a scientific paper?* (Week 9: April 10th)

❖ Checkpoint classes

- Checkpoint 1 (Week 5: March 6th)
- Checkpoint 2 (Week 11: April 24th)

❖ Support classes

❖ Computational environment

- Deucalion

Timeline

❖ **Group formation** (February 11th)

- **MEI**: 4 elements
- **MCA**: 3 elements

❖ **Project selection** (February 13th)

❖ **Tutorials**

- **Tutorial 1**: February 20th
- **Tutorial 2**: March 13th
- **Tutorial 3**: April 10th

❖ **Checkpoint classes**

- Checkpoint 1: March 6th
- Checkpoint 2: April 24th

❖ **Paper submission** (May 25th~~15th~~)

- Conference-like submission
 - 10-page, double-column paper
- Artifact description/evaluation (Code + experiments)
 - GitHub + Zenodo

❖ **Review assignment (MCA only)**

- Bidding (May 26th~~16th~~)
- Reviews (June 8th~~May 29th~~)*

❖ **Presentations** (Week 16: June 1st - 6th)

Team

- ❖ António Sousa (als@di.uminho.pt)
- ❖ André Pereira (ampereira@di.uminho.pt)
- ❖ Cláudia Brito (d13235@di.uminho.pt)
- ❖ Gonçalo Raposo (goncalo.m.raposo@inesctec.pt)
- ❖ Ricardo Macedo (d12010@di.uminho.pt)

Projects

- ❖ Projects are split across 3 main topics
 - *Storage and memory systems*
 - *Energy efficiency and sustainability*
 - *AI-based optimizations for systems*
- ❖ Students can also propose a topic, but its feasibility will be assessed by the supervision team
 - When submitting the topic, include one of the following projects as backup.

Projects

Storage and memory systems

- ❖ Storage system for optimizing distributed AI training in HPC
- ❖ Storage system for optimizing LLM checkpointing in HPC
- ❖ Performance study of key-value stores for HPC
- ❖ FUSE-based file system for small-file workloads in HPC
- ❖ Exploring disaggregated memory for efficient computing

Projects

Energy efficiency and sustainability

- ❖ Characterization of the energy profile of distributed jobs in HPC
- ❖ Carbon footprint of ARM architectures for High Energy Physics processing
- ❖ Carbon-aware strategies for intra- and inter-application scheduling

Projects

AI-based optimizations for systems

- ❖ Adaptive DataLoader configuration for AI training in HPC
- ❖ Adaptive checkpoint frequency control for AI training in HPC
- ❖ Reinforcement learning for adaptive cache configuration

Questions?